FMCG Market Basket Studio

Executive Summary & Technical Architecture

Strategic Objective

To transition traditional market basket analysis from statistical observation to automated margin expansion. By leveraging Expected Commercial Value (ECV) and FP-Growth algorithms, this engine identifies high-profit cross-selling opportunities for E-commerce recommendation systems.

Core Architecture

- **Data Ingestion:** DuckDB engine for out-of-core processing of million-row transaction logs.
- **Algorithmic Engine:** FP-Growth for efficiency, eliminating Apriori's bottleneck.
- **API Layer:** FastAPI microservice delivering JSON responses for real-time checkout.
- **UI Layer:** Streamlit dashboard featuring NetworkX affinity graphs for category managers.

Commercial Valuation Metric (ECV)

The core differentiator of this engine is the monetization of statistical confidence. The Expected Commercial Value is calculated as:

$$ECV = P(\text{Consequent}|\text{Antecedent}) \times Margin_{\text{Consequent}} \times V$$

Where V represents the expected baseline volume (e.g., 1,000 baskets). This allows merchandising teams to prioritize gross margin impact over raw statistical lift.

Deployment Pipeline

The repository enforces strict CI/CD standards, utilizing `uv` for dependency resolution, `ruff` for strict linting, and `pytest` for backend validation. The architecture supports both local parquet file ingestion and enterprise Google BigQuery integrations.